Self-supervised Learning
Self-training, Cycle Consistency & World Models

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Heads up: The next 2 weeks will jump over multiple smaller topics

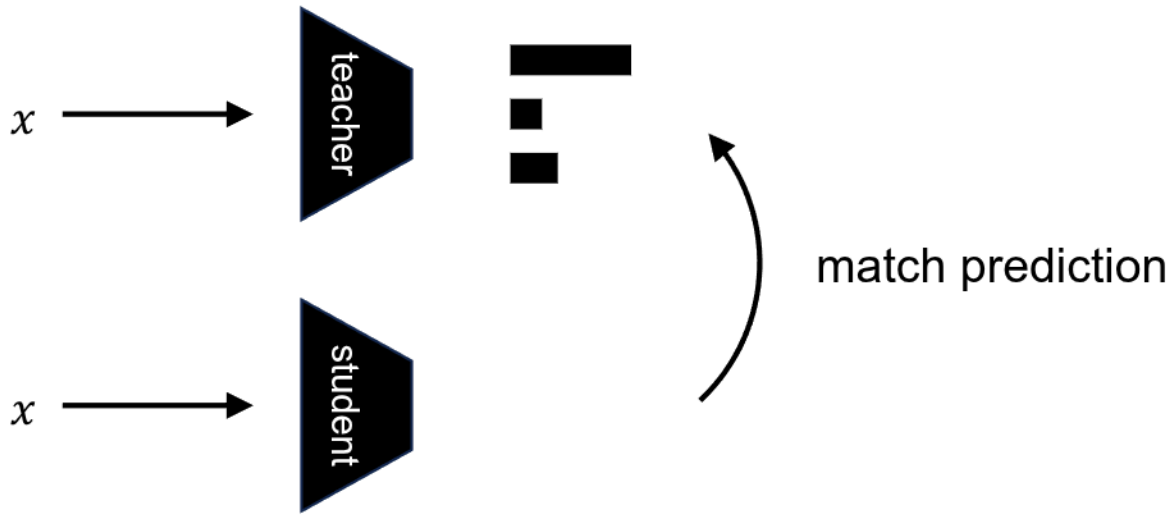
Overcoming data limitations

Multi-crop assumes a dominant visual concept in the image.

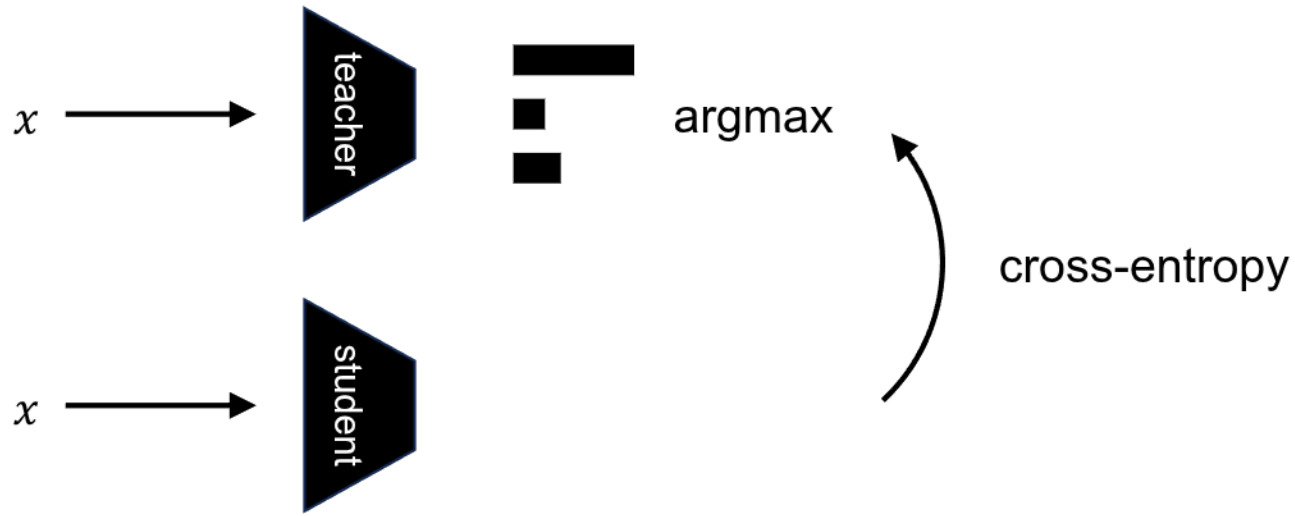
How can we make use of data that does not fit our loss objectives?

- domain transfer
- modality transfer
- scaling to more data

Student-Teacher-Training



A simple way: pseudo labels



The student matches the classification of the teacher, even if the teacher may be uncertain.

Pseudo-labels for semi-supervised learning

Problem: Only few data samples are labelled.

Approach: For unlabelled data, use the network as its own teacher, training on the currently predicted label.

Lee, 2013, „Pseudo Label“

Pseudo-labels for semi-supervised learning

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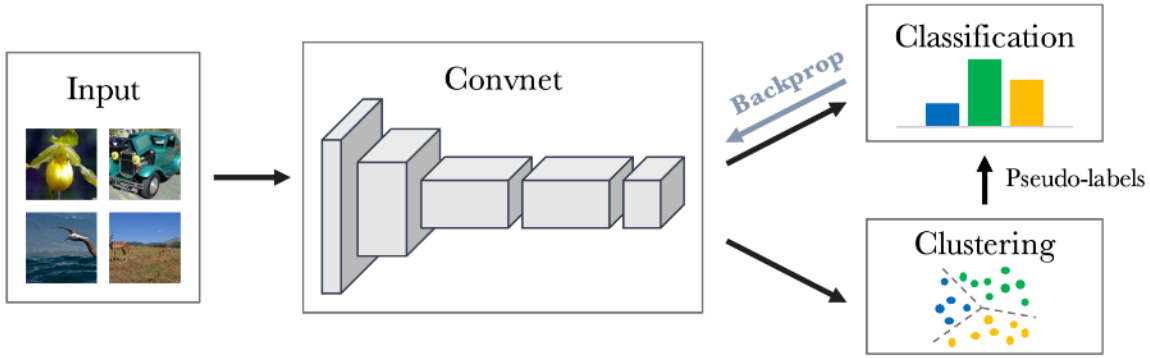
	600 labelled, 59400 unlabelled	1000 labelled, 59000 unlabelled
Supervised Training	91 %	93 %
Supervised + Pseudo Labels	95 %	96 %

MNIST Accuracy

Lee, 2013, „Pseudo Label“

Deep Clustering: Clustering becomes the teacher for the student network

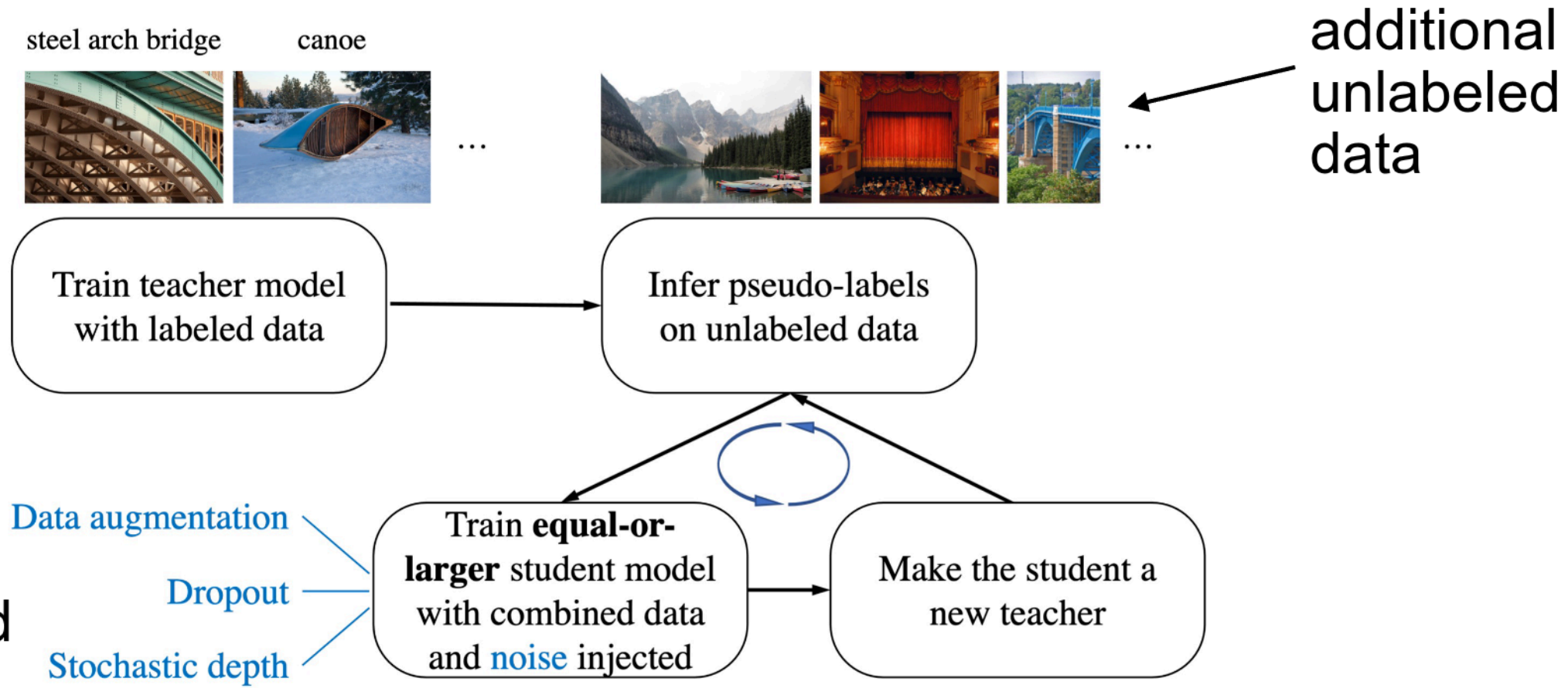


Deep Clustering for Unsupervised Learning of Visual Features

Mathilde Caron, Piotr Bojanowski, Armand Joulin, and Matthijs Douze

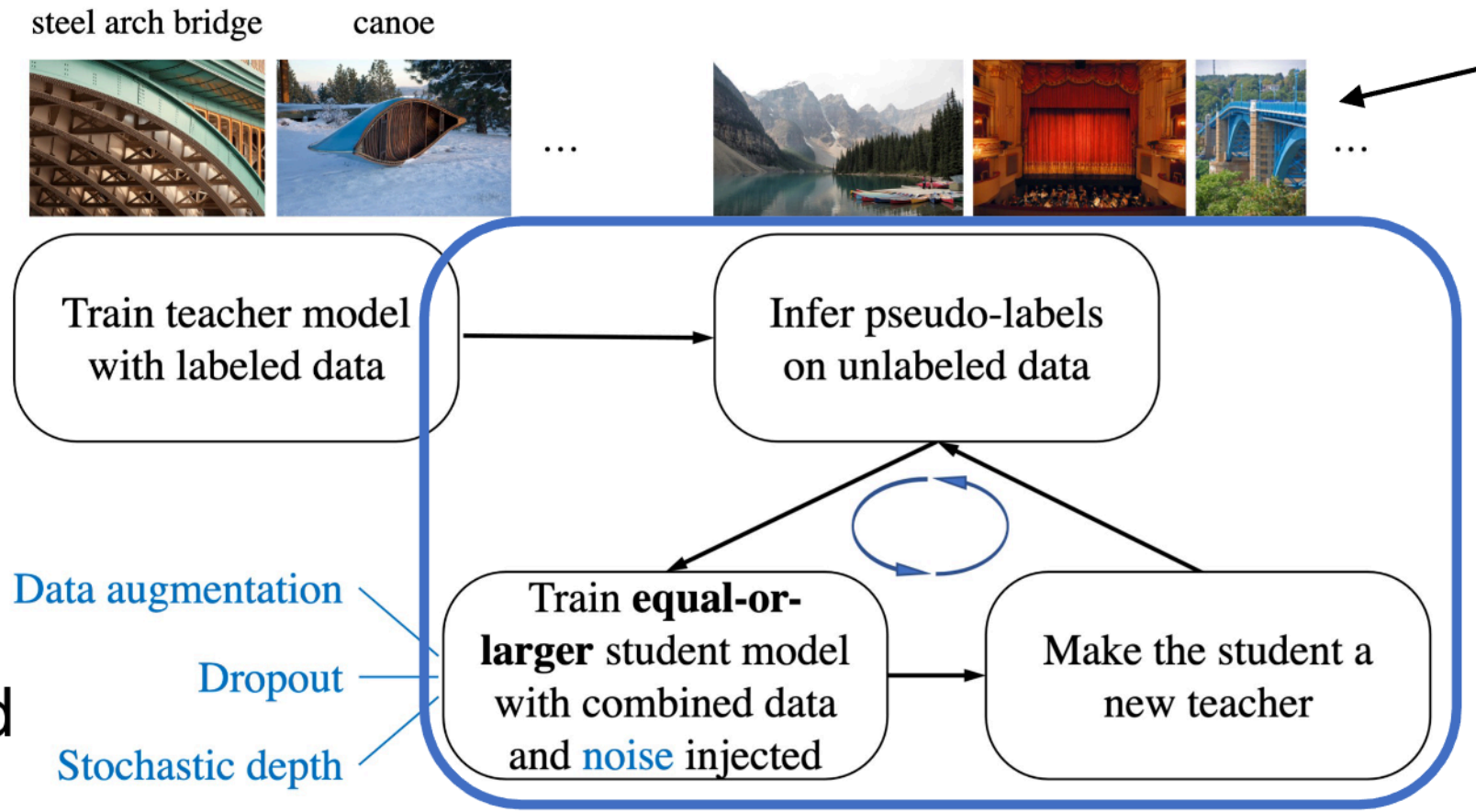
Facebook AI Research
{mathilde,bojanowski,ajoulin,matthijs}@fb.com

Noisy Student: Without more labels, beat SoTA at the time on ImageNet



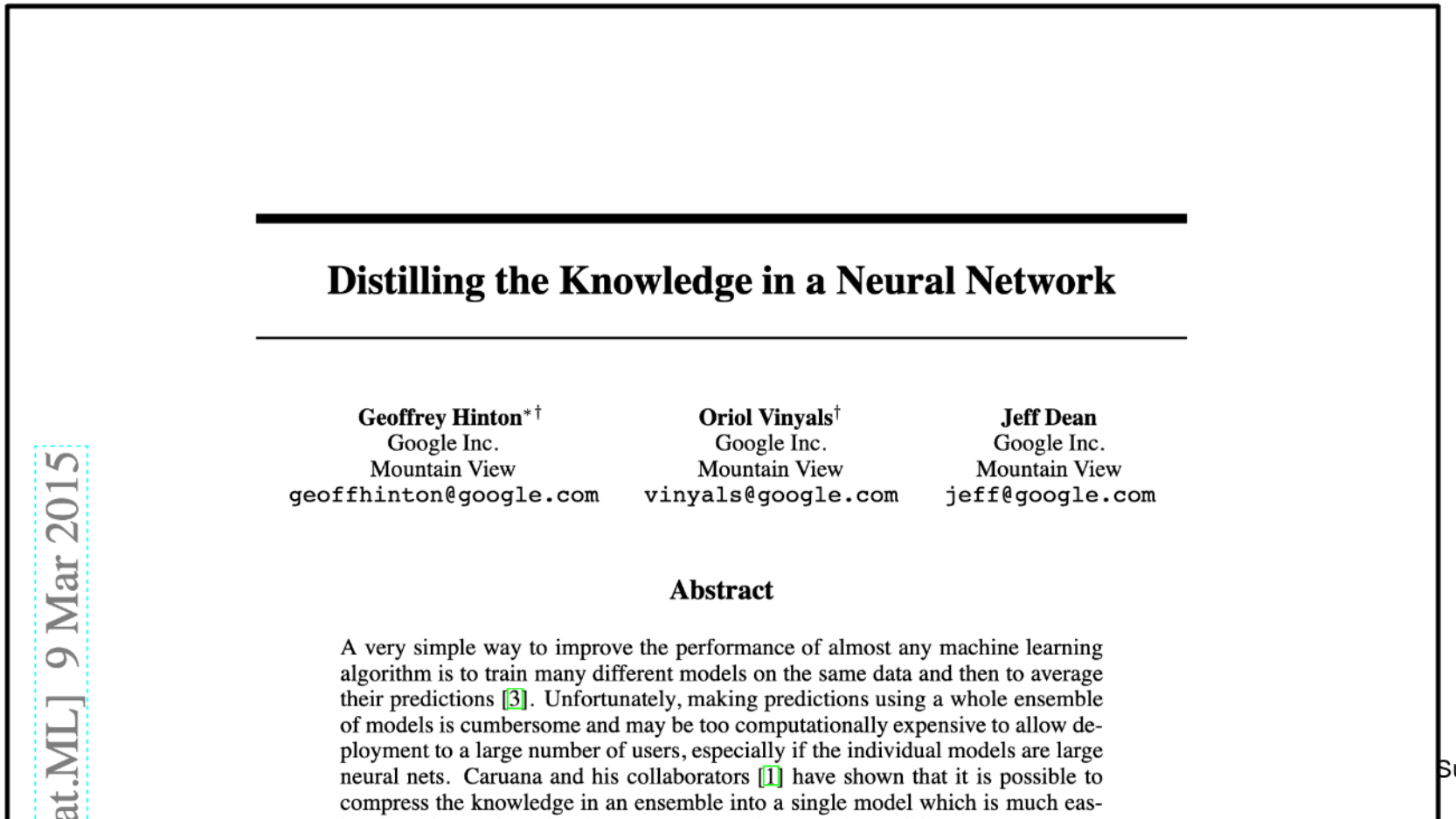
Xie et al, 2020

Noisy Student: Without more labels, beat SoTA at the time on ImageNet



Xie et al, 2020

Distillation



Distilling the Knowledge in a Neural Network

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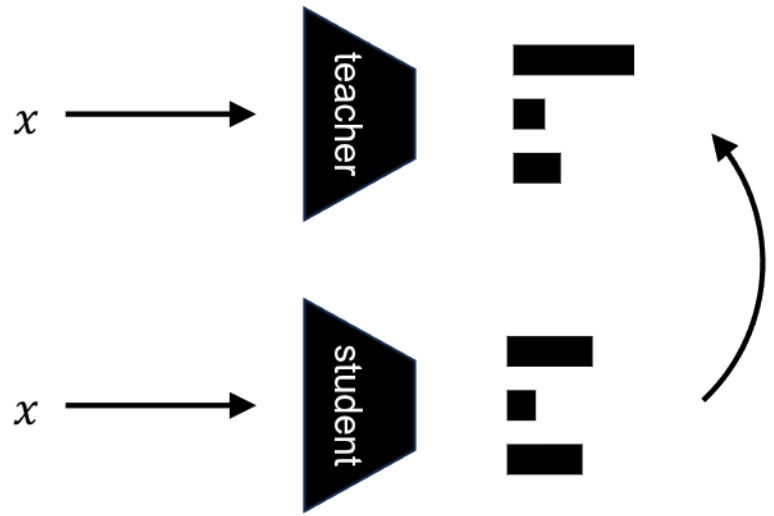
[stat.ML] 9 Mar 2015

Abstract

A very simple way to improve the performance of almost any machine learning algorithm is to train many different models on the same data and then to average their predictions [3]. Unfortunately, making predictions using a whole ensemble of models is cumbersome and may be too computationally expensive to allow deployment to a large number of users, especially if the individual models are large neural nets. Caruana and his collaborators [1] have shown that it is possible to compress the knowledge in an ensemble into a single model which is much eas-

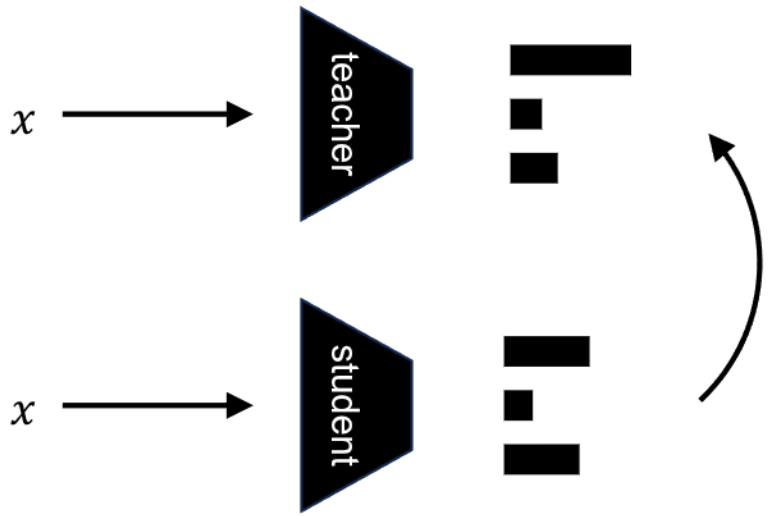
Distillation



cross-entropy between whole distribution
also called „soft target“ or „soft label“

core idea: the output distribution and confidence of the teacher carries important information which the argmax would hide

Distillation

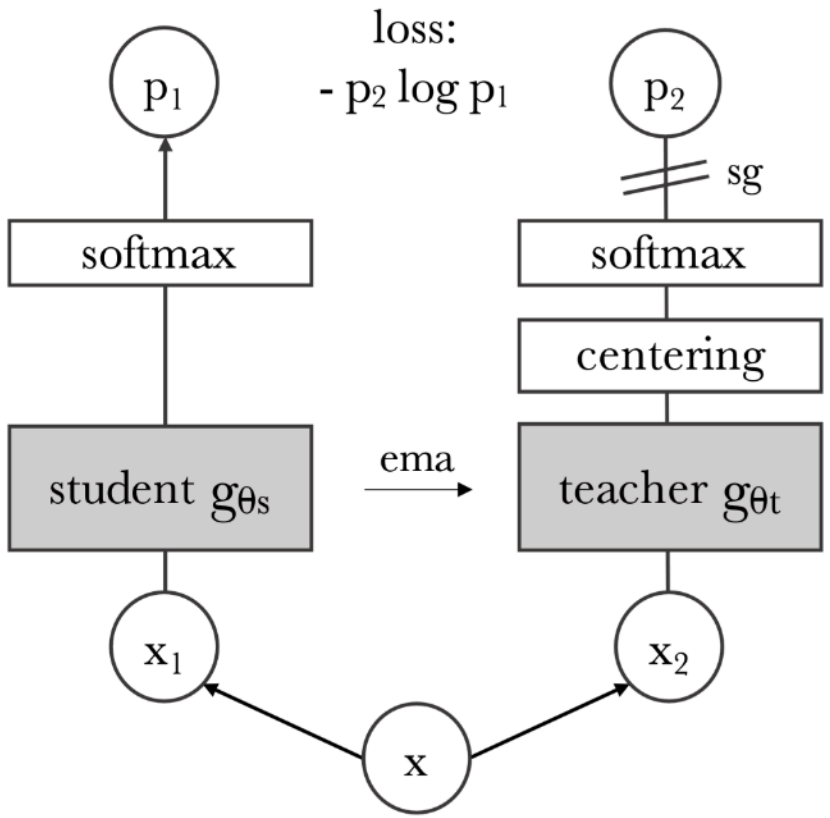


cross-entropy between whole distribution
also called „soft target“ or „soft label“

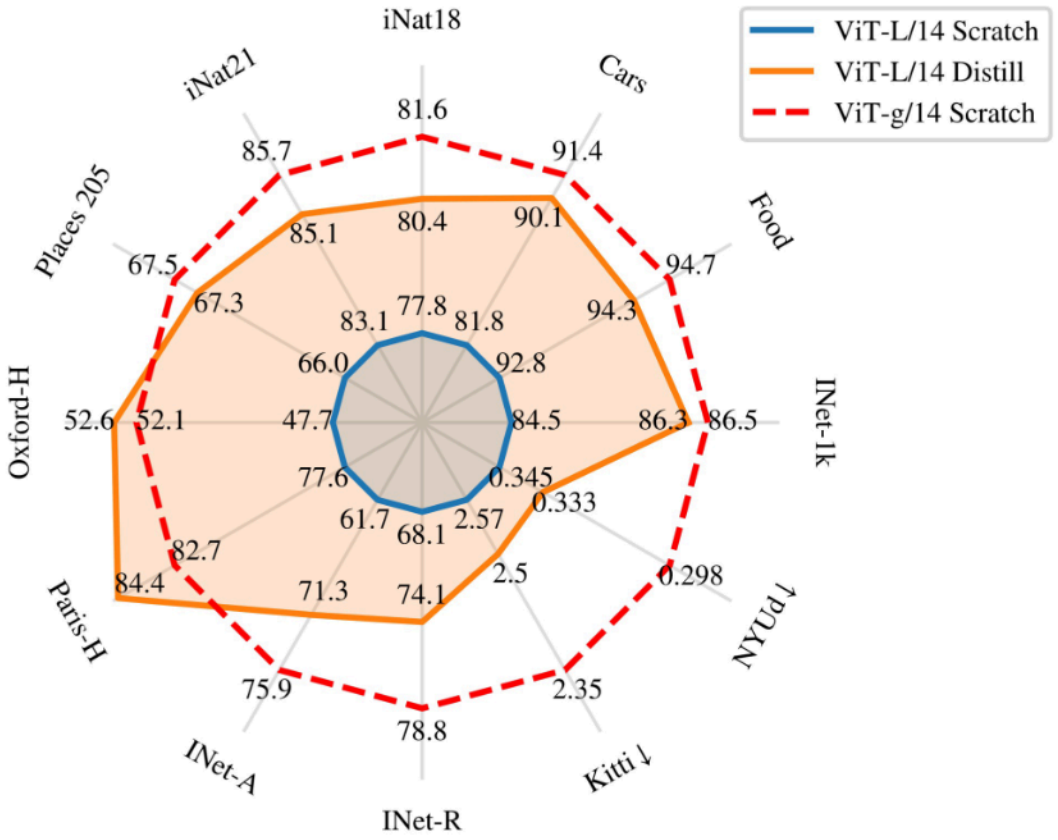
teacher can be

- a single model
- an ensemble of models
- an EMA teacher

DINO = Distillation with no labels



Distillation is often used in model compression



DINOv2 example:

For the same number of parameters, distilling from a larger model achieves far better results than training from scratch with the same data and objective.

Distillation is a classic tool for reducing model size and inference time

Language	Gemini 1.5 Flash	Gemini 1.5 Pro
English	**1.5**	4.3
Japanese	**4.3**	10.9
Chinese	**5.2**	14.1
French	**1.9**	4.7

time [ms] per output character

Gemini

1.5 Flash excels at summarization, chat applications, image and video captioning, data extraction from long documents and tables, and more. This is because it's been trained by 1.5 Pro through a process called "distillation," where the most essential knowledge and skills from a larger model are transferred to a smaller, more efficient model.

Gemini 1.5 Technical Report

DUNE

This CVPR paper is the Open Access version, provided by the Computer Vision Foundation.
Except for this watermark, it is identical to the accepted version;
the final published version of the proceedings is available on IEEE Xplore.

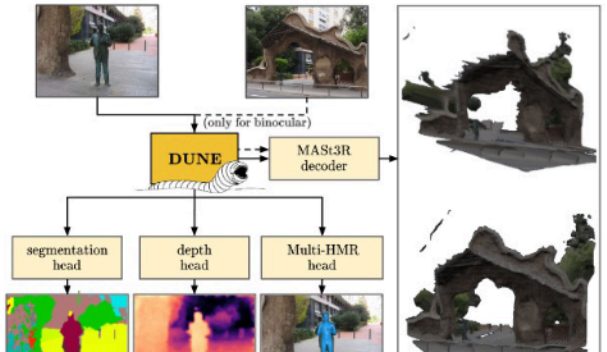
DUNE: Distilling a Universal Encoder from Heterogeneous 2D and 3D Teachers

Mert Bülent Sarıyıldız Philippe Weinzaepfel Thomas Lucas
Pau de Jorge Diane Larlus Yannis Kalantidis
NAVER LABS Europe

https://europe.naverlabs.com/dune

Abstract

Recent multi-teacher distillation methods have unified the encoders of multiple foundation models into a single encoder, achieving competitive performance on core vision tasks like classification, segmentation, and depth estimation. This led us to ask: Could similar success be achieved when the pool of teachers also includes vision models specialized in diverse tasks across both 2D and 3D perception? In this paper, we define and investigate the problem of heterogeneous teacher distillation, or co-distillation—a chal-



DUNE

uses distillation to

- scale up data that is incompatible e.g. with DINO objective
- combine heterogenous teachers that each have different losses and incompatible training requirements
- achieve better generalization over tasks

DUNE

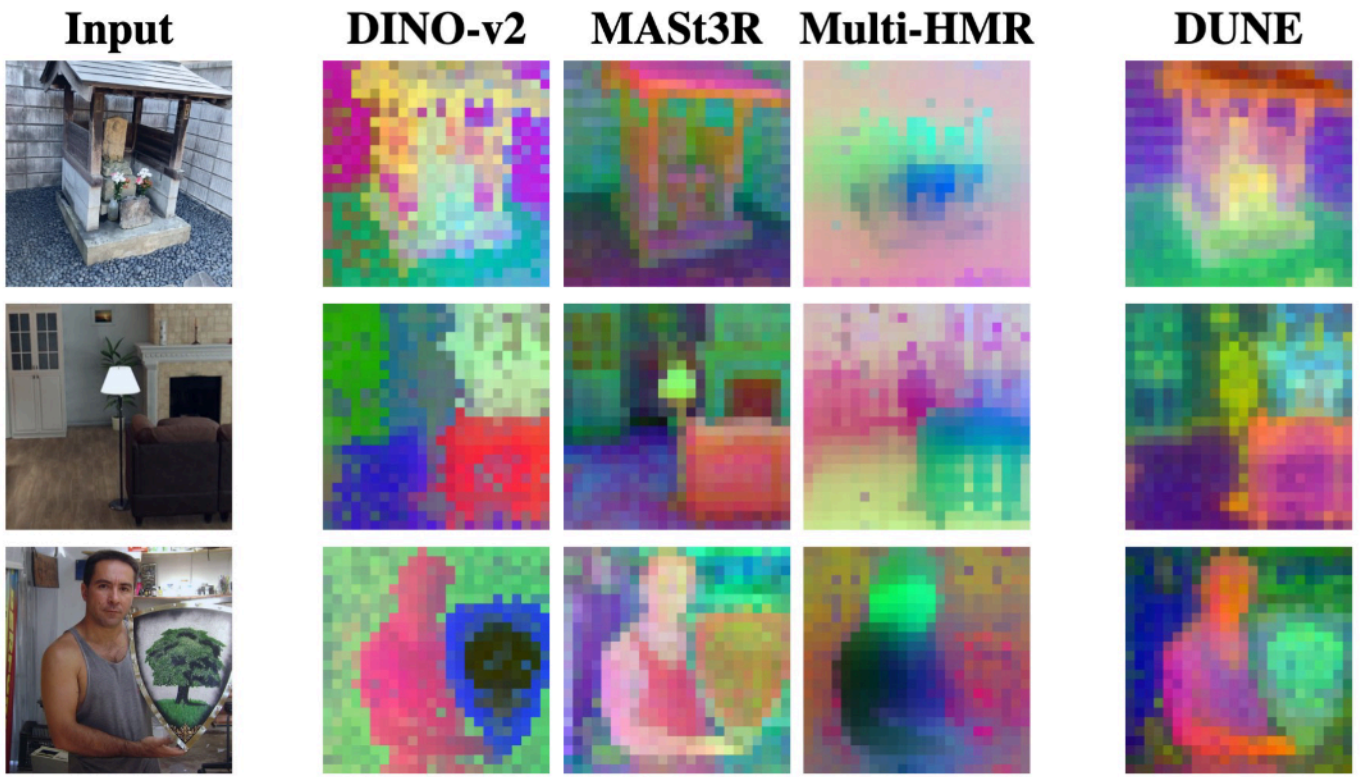
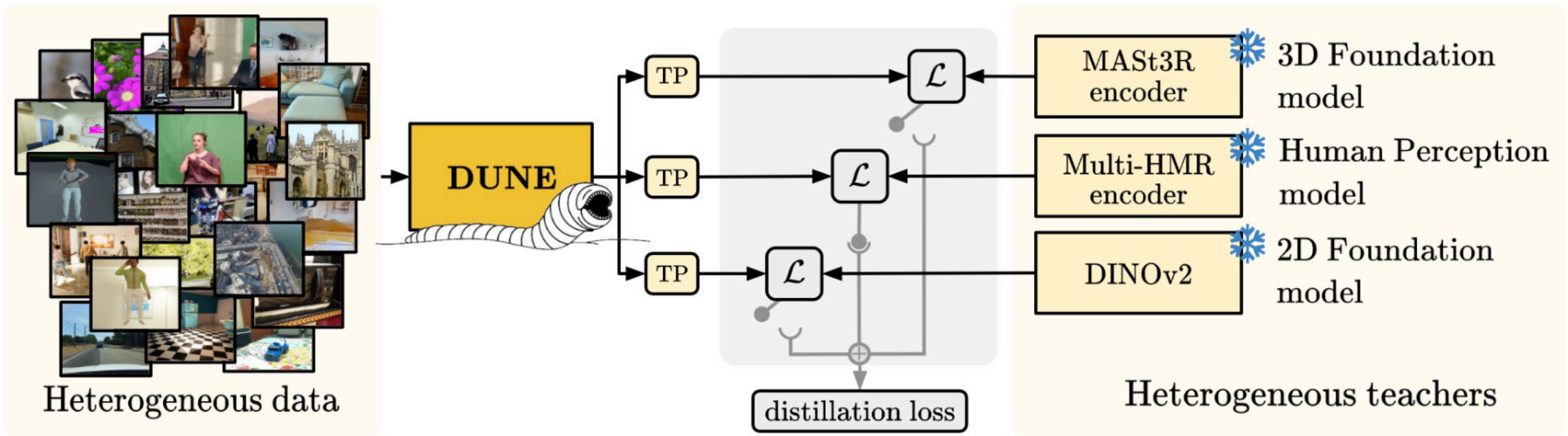
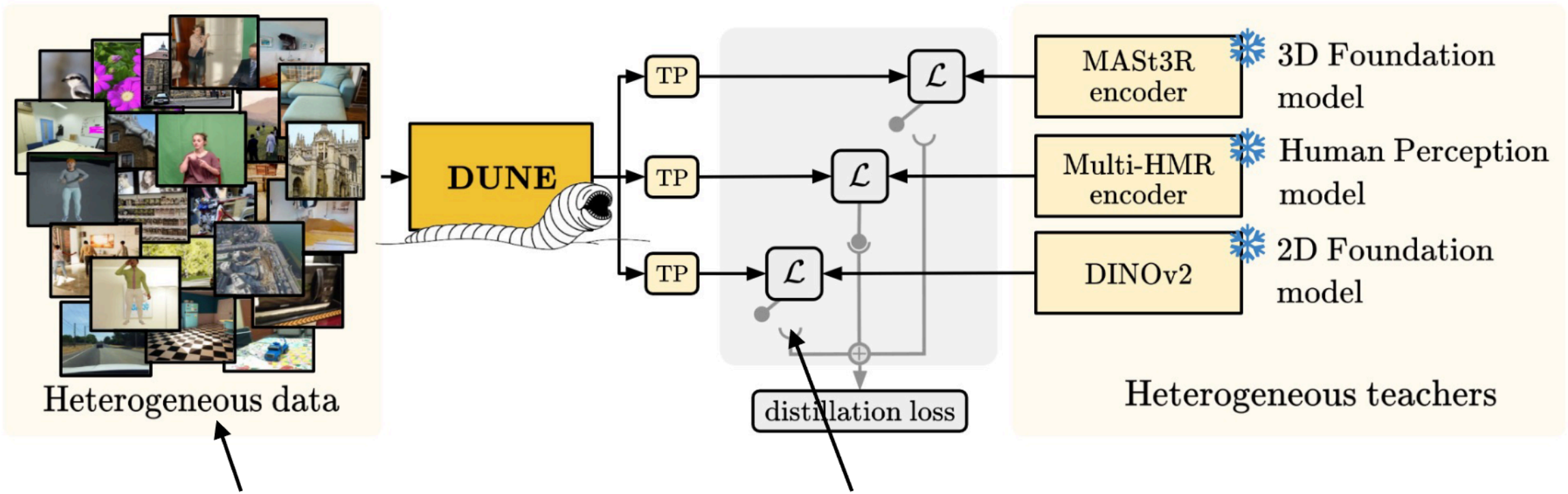


Figure 2. **PCA visualization of encoder outputs.** Given an image, we extract patch embeddings from the encoders of the teacher models and our student, and reduce their dimension to 3 via PCA.

Distilling from Multiple Teachers



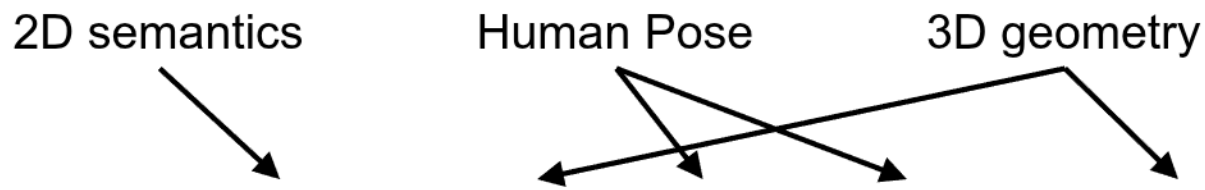
Distilling from Multiple Teachers



for the DINO objective, much of this data would not be compatible

experiments show that this filtering is not necessary. Best option is to train all teachers on all data

Through data scale and teacher correlation, the student can outperform the teachers

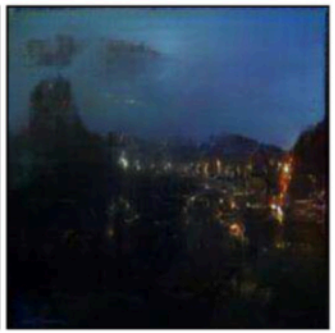


Model	Encoder Arch.	Training Data	Training Res.	ADE20k (mIoU ↑)	NYUd (RMSE ↓)	BEDLAM (F1-score ↑)	BEDLAM (PA-PVE ↓)	MapFree (AUC ↑)
Teacher models								
DINO-v2	ViT-Large	LVD-142M	518	47.7	0.384	-	-	-
Multi-HMR	ViT-Large	HMR-500K	672	-	-	95	36.9	-
MASt3R	ViT-Large	MASt3R-1.7M	512	-	-	-	-	91.2
State-of-the-art ViT-Base encoders								
DINO-v2	ViT-Base	LVD-142M	518	47.3	0.399	86	76.5	89.6
AM-RADIO-v2.5	ViT-Base	DataComp-1B	512	**50.0**	0.718	89	83.2	93.1
DUNE	ViT-Base	DUNE-20.7M	336	44.9	0.377	91	68.3	93.7
DUNE	ViT-Base	DUNE-20.7M	448	45.6	**0.358**	**94**	**56.0**	**94.7**

HARIBO
COLOR-RADO
Für jeden was dabei!
1 kg
Haribo macht Kinder froh und Erwachsene ebenso!

Paired and Unpaired Data

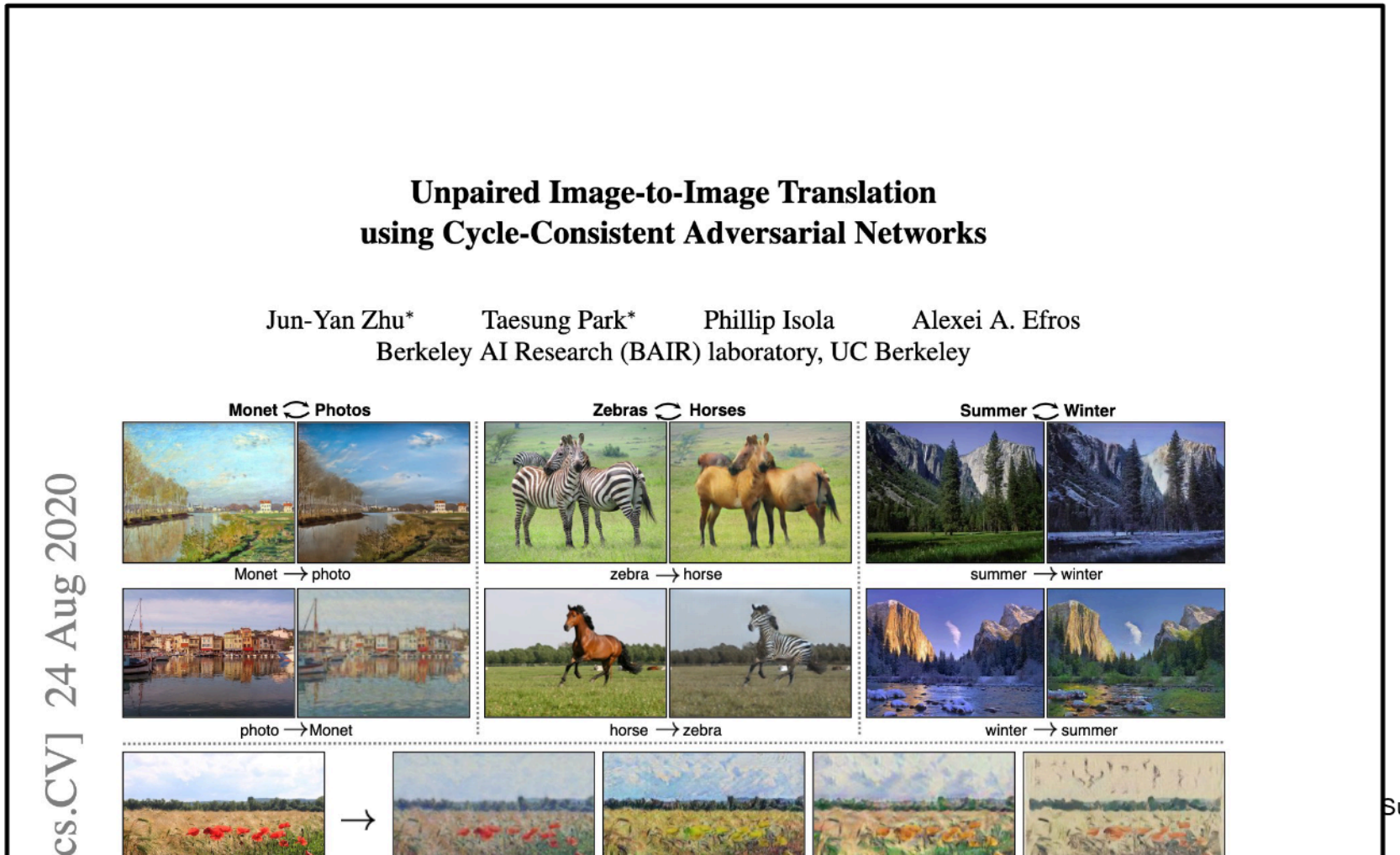
paired data used in conditional generation



strictly, paired data is rather semi-supervised

we might consider them SSL if we can mine pairs without any manual effort, e.g. super-resolution, augmentations

Cycle GAN:
Learning from unpaired data



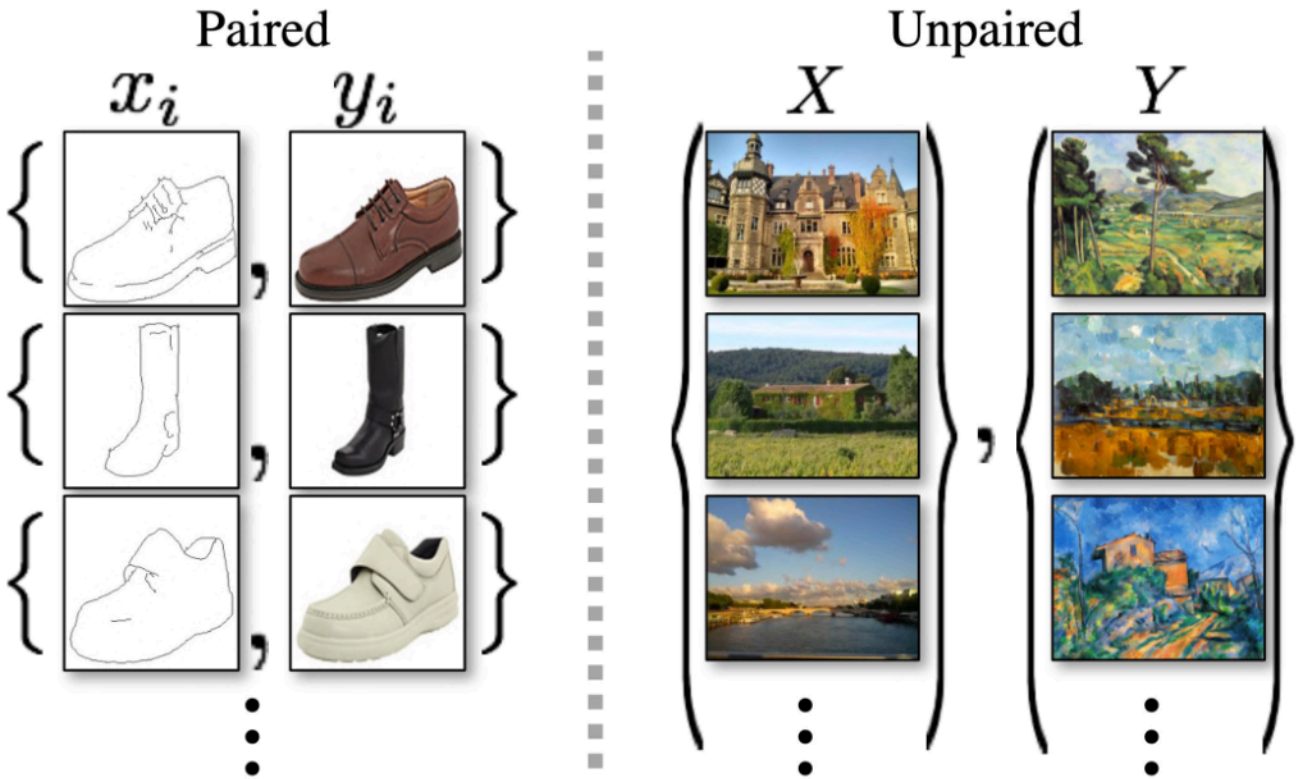
Unpaired Image-to-Image Translation using Cycle-Consistent Adversarial Networks

Jun-Yan Zhu* Taesung Park* Phillip Isola Alexei A. Efros
Berkeley AI Research (BAIR) laboratory, UC Berkeley

cs.CV] 24 Aug 2020

Unpaired data: 2 defined domains, but no direct matches

e.g., it is impossible to have paired matches between Monet paintings and photos



Quick Recap: GAN

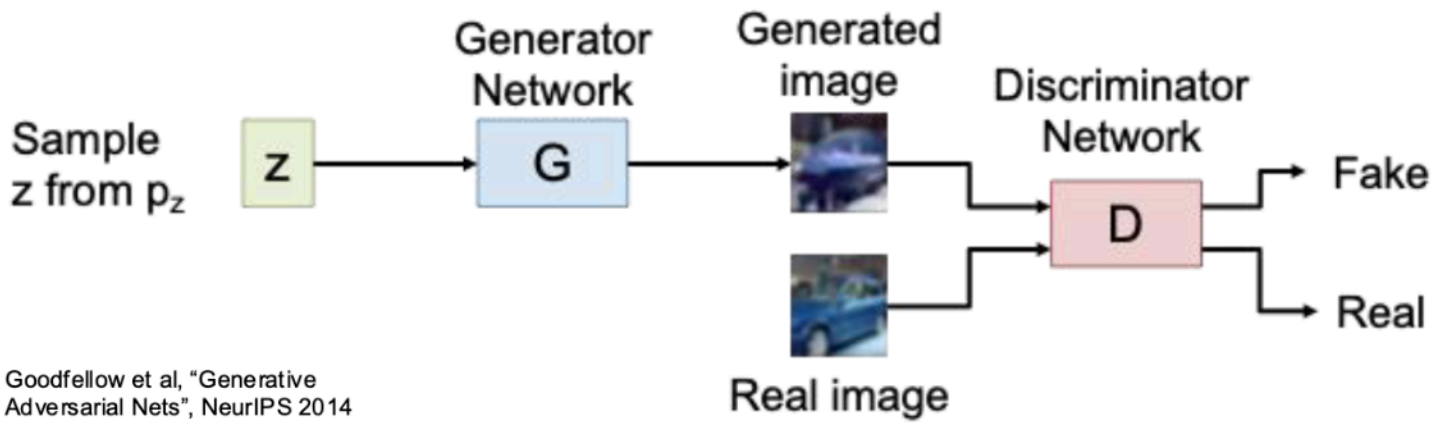
Jointly train generator G and discriminator D with a **minimax game**

This term does not depend on G

Generator wants D(x) = 1 for fake data

Imagine fixing **D**

$$\min_G \max_D \left(E_{x \sim p_{data}} [\log D(x)] + E_{z \sim p(z)} \left[\log \left(1 - D(G(z)) \right) \right] \right)$$

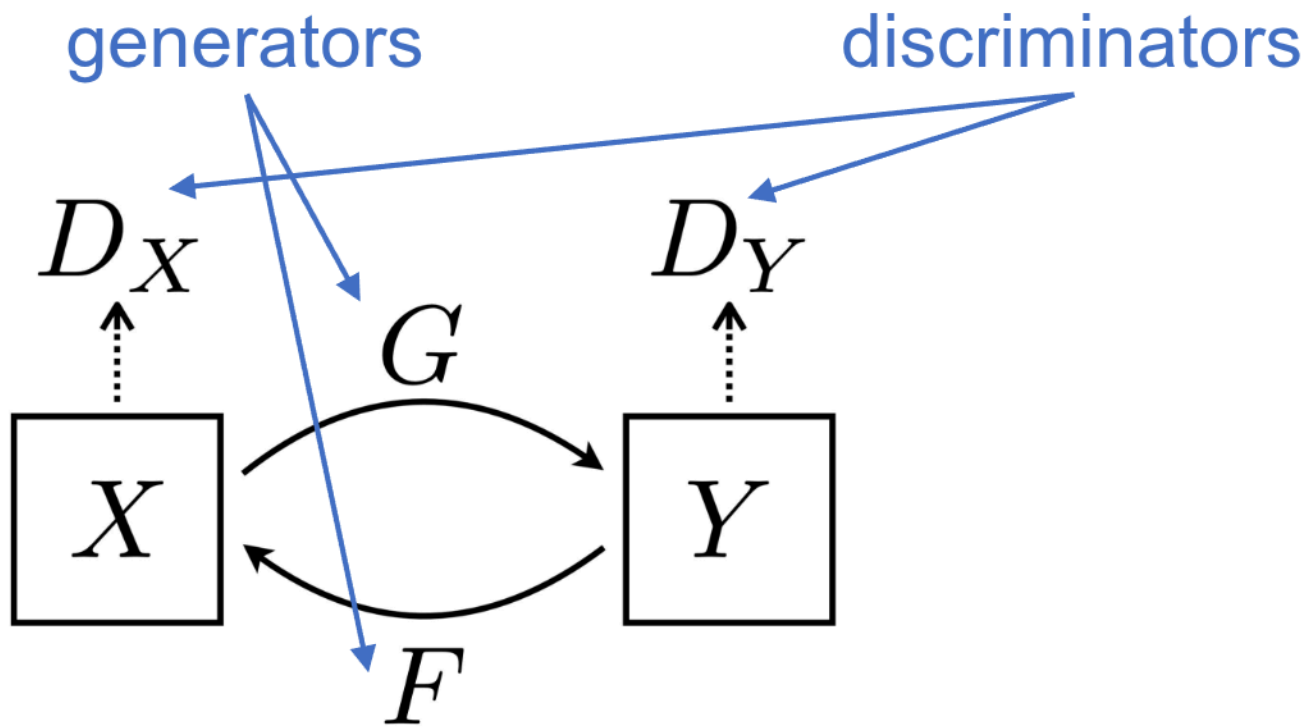


Goodfellow et al, "Generative Adversarial Nets", NeurIPS 2014

D(x) = P(x is real)
D(x) = 0 => fake
D(x) = 1 => real

slide by Li et al, Stanford CS231n

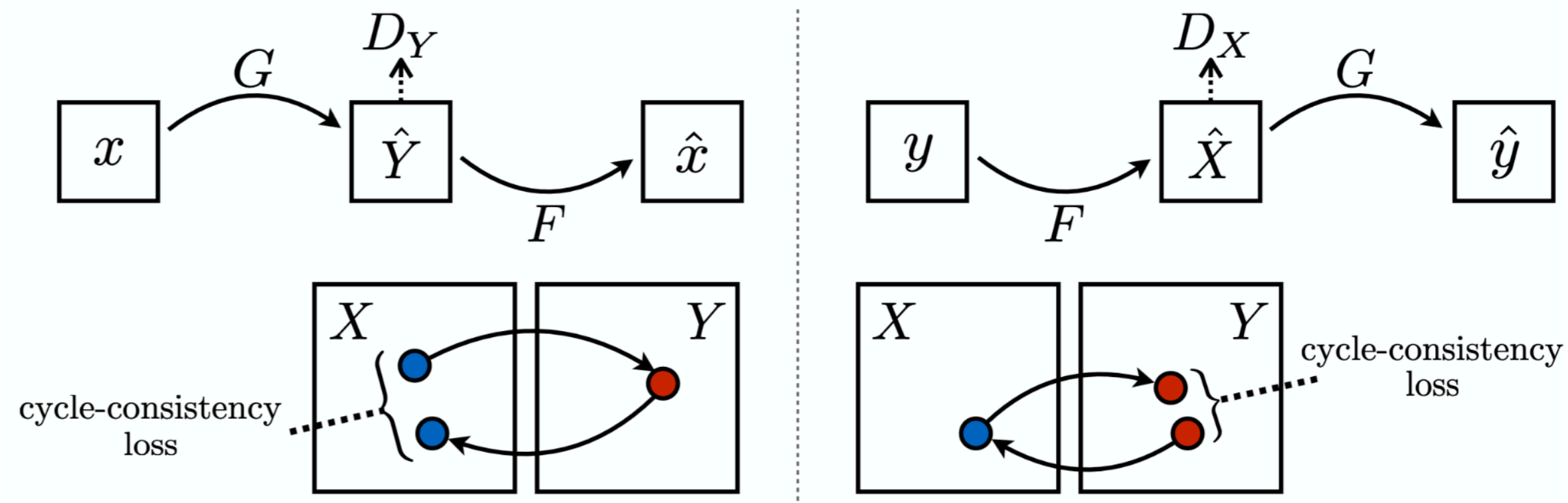
Cycle GAN uses 2 intertwined GANs



Each generator maps from one domain to the other domain.
The discriminators each specialise in one domain.

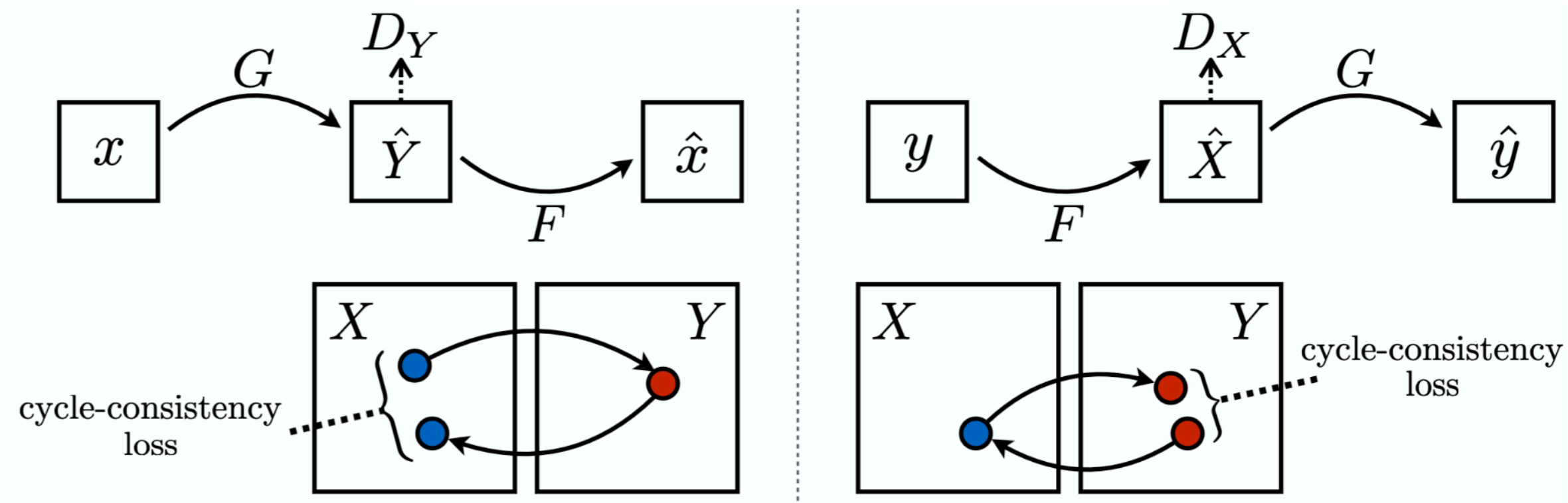
Cycle GAN uses 2 intertwined GANs

for any single image, mapping from one to the other domain and back should return the original image



Cycle GAN uses 2 intertwined GANs

$$\mathcal{L}(G, F, D_X, D_Y) = \mathcal{L}_{\text{GAN}}(G, D_Y, X, Y) + \mathcal{L}_{\text{GAN}}(F, D_X, Y, X) + \lambda \mathcal{L}_{\text{cyc}}(G, F),$$



Monet ⇄ Photos

Monet → photo

photo → Monet

Zebras ⇄ Horses

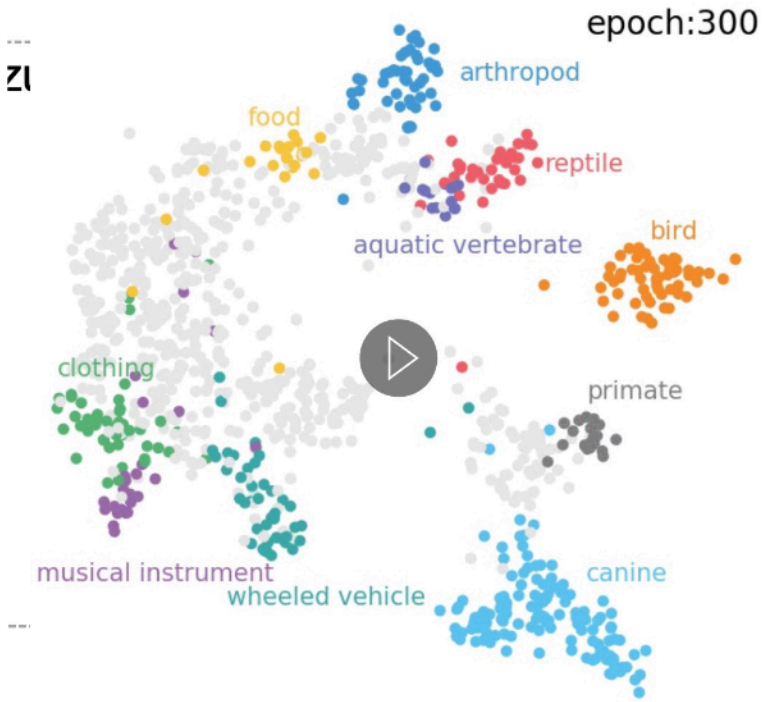
zebra → horse

horse → zebra

HARIBO
COLOR-RADO
Für jeden was dabei!
1 kg
Haribo macht Kinder froh und Erwachsene ebenso!

Feature Extraction with the Goal of Abstraction

Goal: Map from X into a space Z that can be used as an ‚abstraction' of the data that is more convenient to process. E.g., points in Z often have a meaningful distance metric.



Feature Extraction with the Goal of Abstraction

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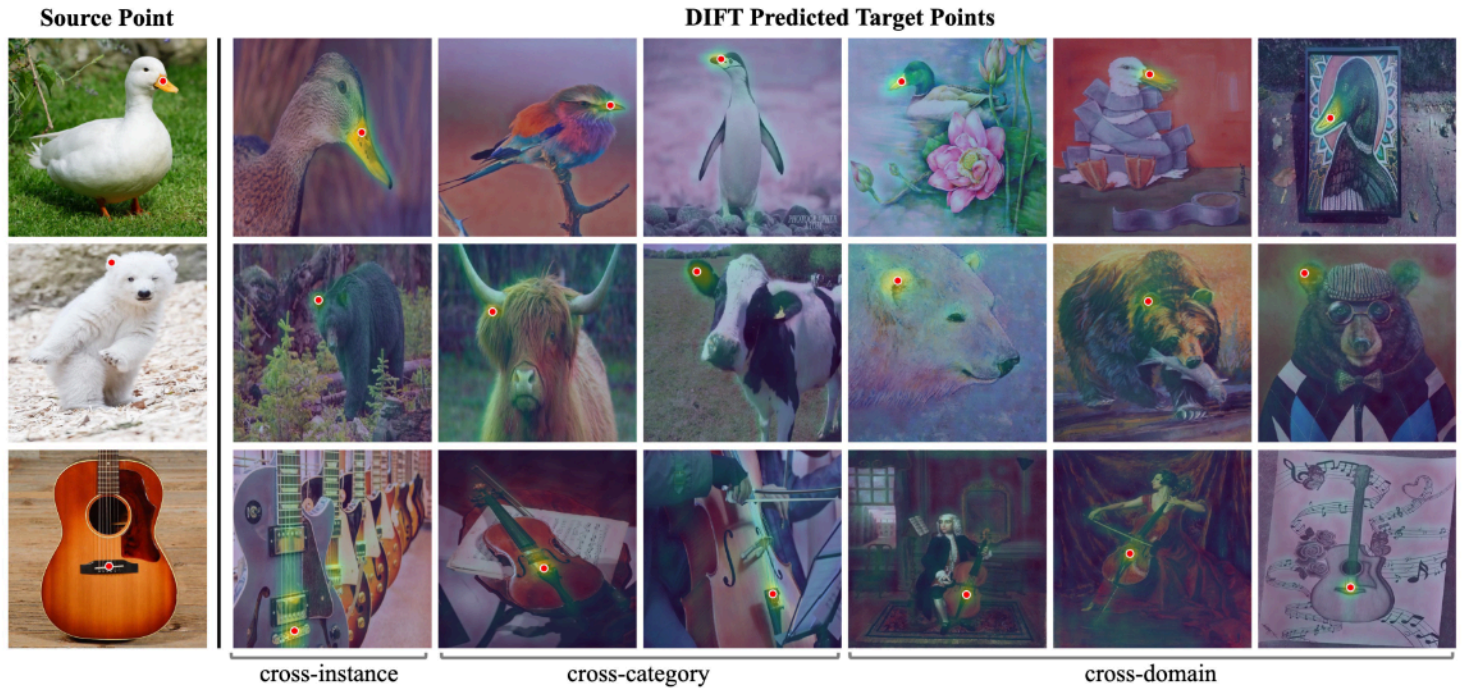
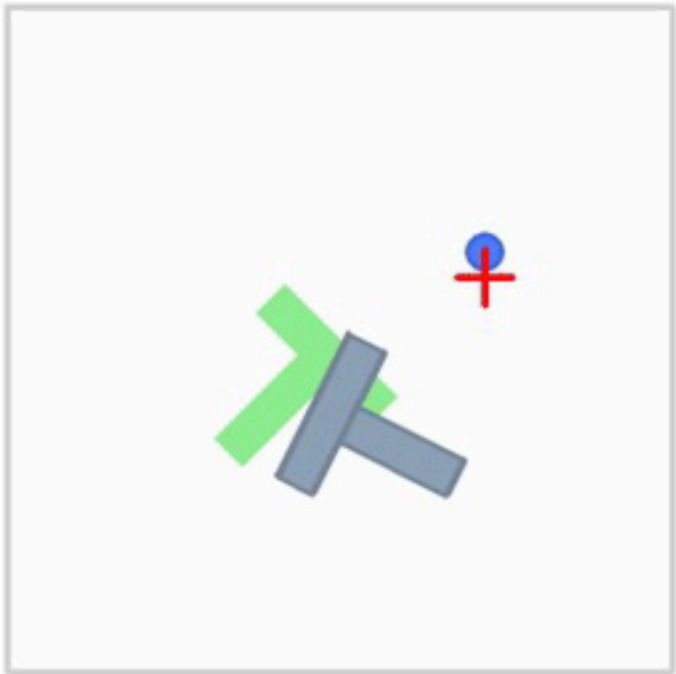


The current frontier for learning abstractions: world models

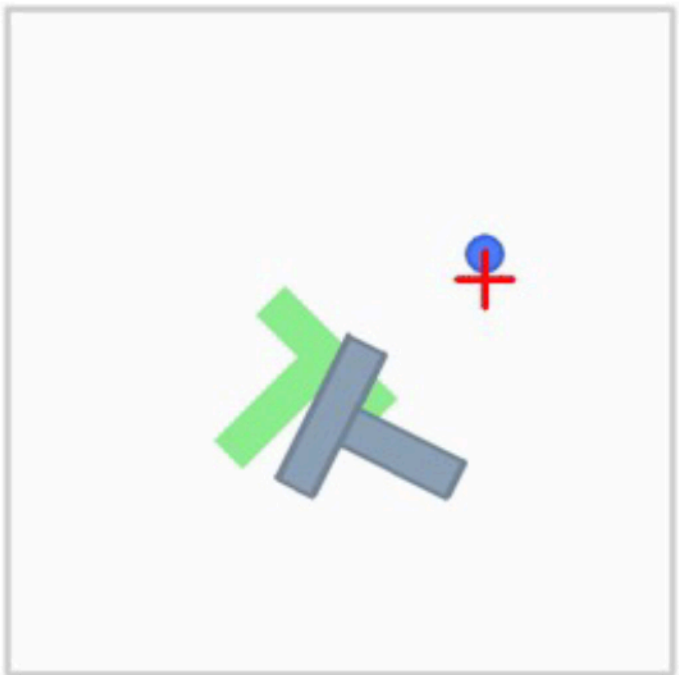


Are world models self-supervised?

An autoregressive model can be a simple world model:

x_0 x_1 ... x_t x_{t+1}

what will happen next?



We can learn this in a self-supervised way from sequential data, such as videos.

Are world models self-supervised?

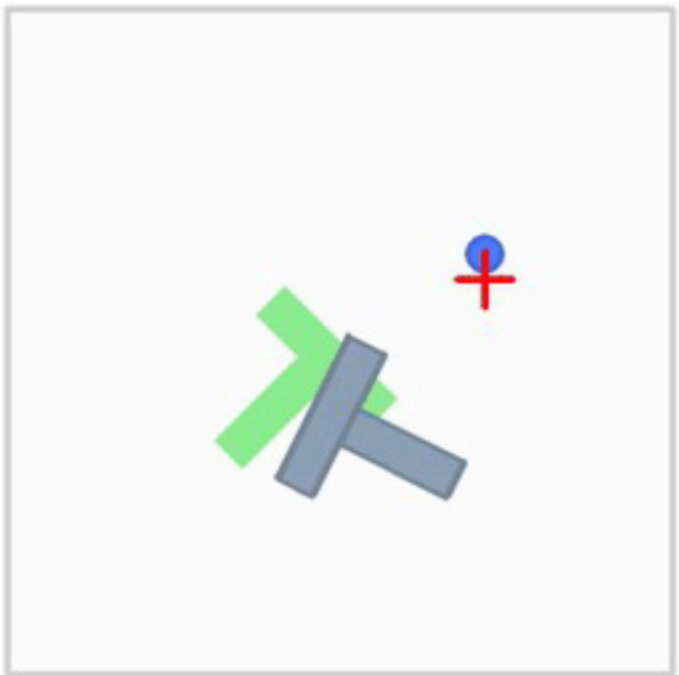
usually we are rather interested in action-conditioning:

x_0 x_1 … x_t

a_0 a_1 … a_t

x_{t+1}

what will happen next?



Either we need to record paired data (not SSL) or we can extract the action from the raw data (we may call this SSL)

DINO-WM

[cs.RO] 1 Feb 2025

DINO-WM: World Models on Pre-trained Visual Features enable Zero-shot Planning

Gaoyue Zhou [1] **Hengkai Pan** [1] **Yann LeCun** [1 2] **Lerrel Pinto** [1]

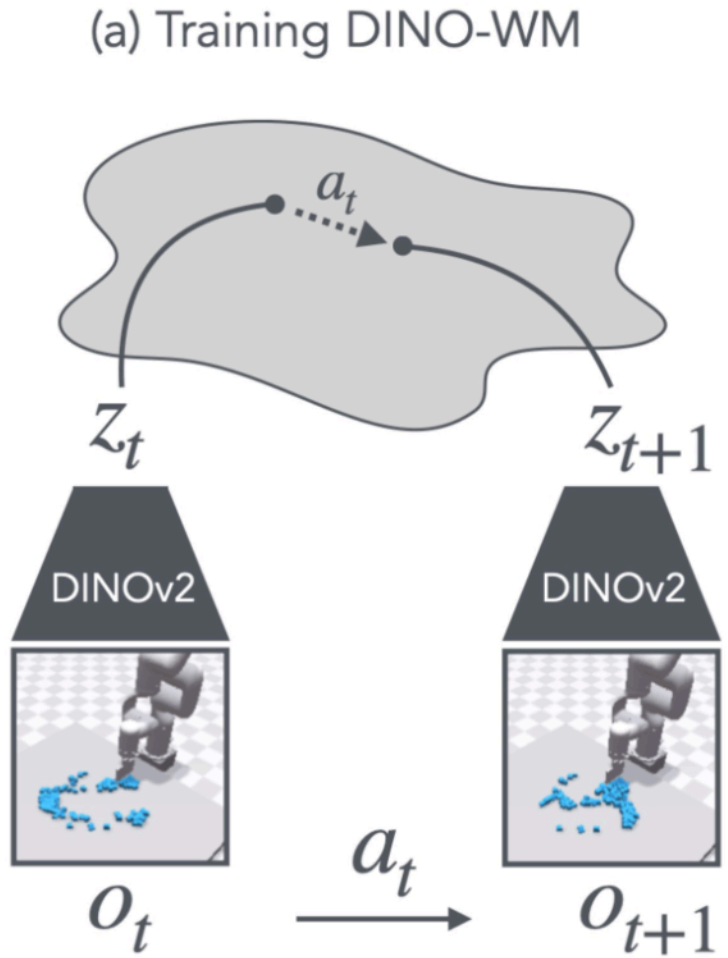
Abstract

The ability to predict future outcomes given control actions is fundamental for physical reasoning. However, such predictive models, often called world models, remains challenging to learn and are typically developed for task-specific solutions with online policy learning. To unlock world models' true potential, we argue that they should 1) be trainable on offline, pre-collected trajectories, 2) support test-time behavior optimization, and 3) facilitate task-agnostic reasoning. To this end, we present DINO World Model (DINO-WM), a new method to model visual dynamics without reconstructing the visual world. DINO-WM leverages spatial patch features pre-trained with DI-

2024). Despite this progress, generalization remains a major challenge (Zhou et al., 2023). Existing approaches predominantly rely on policies that, once trained, operate in a feed-forward manner during deployment—mapping observations to actions without any further optimization or reasoning. Under this framework, successful generalization inherently requires agents to possess solutions to all possible tasks and scenarios once training is complete, which is only possible if the agent has seen similar scenarios during training (Reed et al., 2022; Brohan et al., 2023b;a; Etukuru et al., 2024). However, it is neither feasible nor efficient to learn solutions for all potential tasks and environments in advance.

Instead of learning the solutions to all possible tasks during training, an alternate is to fit a dynamics model on

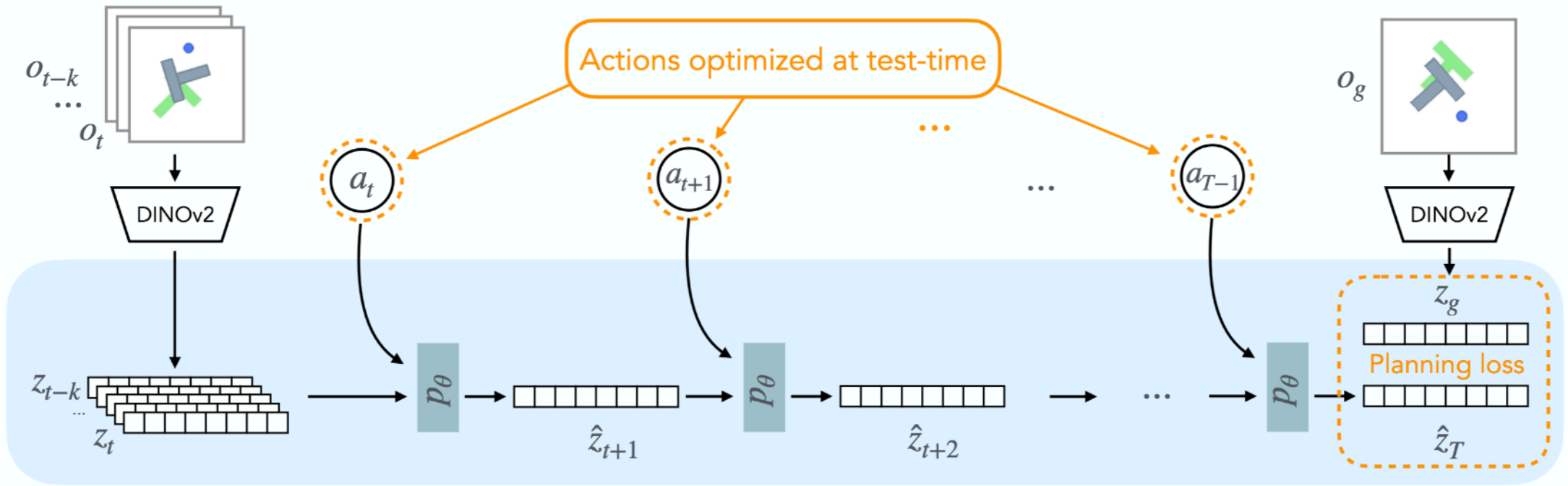
DINO-WM uses the DINO features as a prediction space.



Train to predict the encoding of the next frame:

$$\mathcal{L}_{pred} = \|p_\theta\left(\text{enc}_\theta(o_{t-H:t}), \phi(a_{t-H:t})\right) - \text{enc}_\theta\left(o_{t+1}\right)\|^2$$

To plan, we search over actions to arrive at the encoding of the goal.



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